

The Ambiguous Band: Latency-Safe Quantum Kernel Re-ranking for Credit Card Fraud Detection

A hybrid classical–quantum architecture that places quantum computation where it is both tractable and valuable — and nowhere near the authorisation path.

Challenge: HSBC — Quantum-Enhanced Credit Card Fraud Detection

Team: Entangled Minds

Paradigm: Hybrid classical–quantum, gate-based kernel method

Summary. A tuned gradient-boosted baseline already resolves the overwhelming majority of card-not-present transactions correctly. Its residual error — and essentially all of the false-positive cost the challenge statement quantifies — is concentrated in a thin *ambiguous band* of scores, typically 2–5% of volume. We propose **QBR (Quantum Band Re-ranking)**: a quantum kernel model trained only on that band, converted into an explicit finite-dimensional feature vector by a **Landmark Quantum Kernel Embedding (LQKE)**, and consumed by a calibrated classical head. Three properties follow directly from this design. The band is small, so the $O(n^2)$ cost of quantum kernel estimation becomes tractable — the class imbalance that handicaps classical training is precisely what makes quantum feasible here. The embedding is explicit and finite, so SHAP and prototype-based attribution work natively, satisfying model-governance review. And no QPU call sits in the 100–300 ms authorisation path, so the latency constraint is met by construction rather than by hope. We commit in advance to a matched-protocol evaluation against four controls, and to publishing a negative result if quantum does not clear them.

1. Problem Framing

The enterprise problem is not classification accuracy; it is the detection–experience tradeoff. The challenge statement's own figures make this unambiguous: global card fraud losses were \$34bn in 2023, while *falsely declined legitimate transactions* ran to \$443bn — an order of magnitude larger — with 40%+ of customers abandoning a purchase after a decline, and up to 95% of transaction-monitoring alerts proving false. For an issuing bank, the marginal pound is not won by catching more fraud in general. It is won by removing false positives *without* surrendering recall. Any proposal that reports only a headline AUC has not engaged with the actual economics.

Where does a 0.9459-AUC ensemble actually fail? Not on the easy mass of transactions. A well-tuned LightGBM assigns confident, correct scores to the large majority of traffic; its errors cluster in a narrow score interval where the evidence genuinely conflicts — a plausible amount at a plausible merchant from a device with a partially matching fingerprint. We call this interval the **ambiguous band**, $\tau_{\text{low}} < s(x) < \tau_{\text{high}}$. Characterising this band quantitatively is the first deliverable of our PoC, because the entire proposal rests on the claim that it is small in volume and dense in error. Published fraud-model operating characteristics and the imbalance profile of both target datasets make a band of 2–5% of volume containing the substantial majority of both false positives and false negatives the expected regime; we treat the exact figure as a measurement, not an assumption.

Why quantum methods offer a credible edge in this specific region — and only here. Three arguments, each of which also explains why we do *not* propose quantum for the full pipeline:

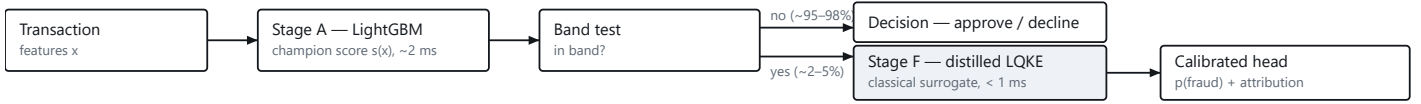
- **Tractability follows from rarity.** Quantum kernel methods require $O(n^2)$ circuit evaluations and are hopeless on 590,540 transactions. Restricted to the ambiguous band — and further to its landmark set — n falls to the low thousands and the landmark Gram matrix to $m \approx 64\text{--}256$. The extreme class imbalance the statement lists as bottleneck #3 is, under this architecture, the enabling condition rather than the obstacle.
- **Inductive-bias complementarity.** Gradient-boosted trees partition space with axis-aligned splits and are exceptionally efficient at low-order, threshold-like interactions — which is why they win this problem class. Their known weakness, named in the statement as "limited kernel expressiveness", is smooth high-order interaction among many weakly-informative continuous features. That is exactly the structure of the Vesta V1–V339 block and the ULB PCA components, and exactly what a quantum feature map produces without manual engineering. We are not asking quantum to beat trees at what trees do well; we are asking it to supply the representation trees cannot reach, in the one region where trees have run out of signal.
- **The residual is where headroom lives.** The statement itself notes "headroom, particularly in capturing subtle cross-feature patterns while minimising customer friction." Concentrating an expensive, expressive model on 3% of volume is standard cost-benefit engineering; it is also the only deployment shape in which a near-term QPU is economically defensible.

What we explicitly do not claim. We claim no asymptotic quantum speed-up and no complexity-theoretic separation. We claim a measurable, auditable, statistically defensible change in AUPRC and in false positives at fixed recall, attributable to the quantum component *after* controlling for architecture and dimensionality — or a clean, published negative result. The challenge statement asks teams to "characterize under what conditions quantum approaches perform differently"; our design makes that the primary scientific output rather than a secondary one.

2. Technical Approach

Paradigm: hybrid classical–quantum; gate-based; kernel-based (non-variational) at inference, with an optional variationally-trained feature map. Quantum execution is confined to offline training; inference is classical arithmetic.

ONLINE — authorisation path, < 300 ms total budget



OFFLINE — training, no latency constraint

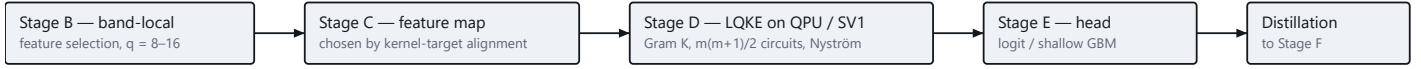


Figure 1. QBR architecture. Shaded blocks involve quantum-derived computation. Every QPU invocation occurs below the dashed line, during training. The online path contains no quantum call and no kernel evaluation against landmarks — only the distilled surrogate.

Stage A — classical gate and band definition. A tuned LightGBM/XGBoost champion, trained per the public IEEE-CIS state of the art, scores all traffic. Thresholds τ_{low} , τ_{high} are fitted on a validation fold to the smallest band capturing a target share (we propose $\geq 75\%$) of the champion's residual errors. Band width is a tunable operating decision, and we will report the full band-width/lift curve rather than a single point.

Stage B — band-local feature selection. Encoding hundreds of features into a circuit is not practical, as the statement notes. We reduce to $q = 8\text{--}16$ features using mutual information plus SHAP importances computed *within the band*. This distinction matters: the features that separate fraud among ambiguous cases are not the features that dominate globally, and band-local selection is itself a reportable finding.

Stage C — feature map. Candidates: ZZ/Pauli two-design (per Karimi et al., the best-documented map for this dataset family), dense-angle encoding (per the quantum-autoencoder anomaly-detection literature), and data re-uploading. Selection is made on validation folds by *kernel-target alignment*, never by test performance — a pre-committed protocol point that prevents the map-shopping that inflates much of the published QML literature.

Stage D — Landmark Quantum Kernel Embedding. We select m landmarks by k-means++ over the band, stratified to over-represent fraud. The QPU (or SV1) computes only the $m \times m$ landmark Gram matrix $K = m(m+1)/2$ circuit evaluations, independent of dataset size. A Nyström construction $z(x) = \Lambda^{-1/2} U^T [k(x, \ell_1), \dots, k(x, \ell_m)]$ then yields an **explicit, finite-dimensional, classical feature vector**. This single step is what converts an opaque kernel machine into something a bank's model-risk function can actually review.

Stage E — calibrated classical head. A regularised logistic regression or shallow GBM over [band features $\oplus z(x)$], probability-calibrated by isotonic regression so the output is a true probability suitable for threshold-setting and economic optimisation.

Stage F — distillation, and the objection it must survive. Computing $z(x)$ exactly requires m kernel evaluations per transaction — acceptable for batch review, not for live authorisation. We therefore train a small classical surrogate to regress $z(x)$ from raw features, and deploy the surrogate. The obvious objection is fatal if unaddressed: *if a classical model can reproduce the quantum embedding, the quantum step was unnecessary*. Our position is that this conflates two different learning problems. The surrogate learns $z(x)$ under dense, continuous, per-example supervision; discovering the same representation from a sparse binary label at 0.172–3.5% positive rate is a far harder problem, and is what classical pipelines demonstrably have not done. If both the fidelity (R^2 of surrogate against true z) and the end-to-end lift survive distillation, the correct interpretation is that quantum acted as a **representation-discovery** mechanism amortised into model weights, not as a runtime necessity — a weaker but genuine and commercially far more deployable claim. We treat this as the central open question of the PoC and will report the fidelity/lift decomposition either way.

3. Feasibility and Resource Requirements

Quantum resources. Width $q = 8\text{--}16$ qubits, sitting comfortably within IQM Garnet (20q), IonQ Aria (25q), Forte Enterprise (36q) and IQM Emerald (54q) on Amazon Braket. Circuit depth for a second-order Pauli map at this width is modest and shallow enough for current two-qubit fidelities (99.5% on Garnet/Emerald). The decisive property is that **hardware cost scales as $m^2/2$, independent of the 590,540-row dataset:**

Landmark set	Circuit pairs, $m(m+1)/2$	Shots @ 500/circuit	Intended venue
$m = 256$ (full embedding)	32,896	16.4 M	SV1 / local cuQuantum
$m = 128$	8,256	4.1 M	SV1, DM1 noise model
$m = 64$	2,080	1.04 M	QPU — primary hardware run
$m = 32$ (hardware replication)	528	0.26 M	QPU — cross-device check

Per-shot and per-task pricing differs by an order of magnitude across Braket devices; we will size m and shot count to a stated hardware budget cap agreed at Phase 2 kickoff rather than commit to a device now. The circuit arithmetic above is fixed and auditable; the device choice is a budget decision made against it. Bulk experimentation runs on SV1; hardware is used for validation, not for bulk training.

Stratified subsampling, stated explicitly. As the statement requires, all quantum-executed subsets are drawn by stratified sampling preserving the fraud/non-fraud ratio of the source partition, and every submission will state the exact sample count used for quantum execution. Landmark selection is stratified to over-represent fraud, and this deviation from the base rate is reported as a design parameter, not concealed.

Classical resources. A single GPU workstation or one g5.2xlarge-class instance suffices for baseline training, SHAP, the ablation ladder and the surrogate. Local statevector simulation via cuQuantum/PennyLane covers rapid iteration at $q \leq 16$.

Data. IEEE-CIS (primary; 590,540 train rows, up to 393 features, 3.5% fraud, natural TransactionDT temporal ordering) and the ULB European Cardholder set (secondary; 284,807 rows, 0.172% fraud, ODbL) — two datasets, per the statement's guidance. Sparkov is held in reserve for a controlled scaling experiment only. **No HSBC proprietary data is required at any point**, which removes the data-access dependency that typically consumes the first month of a bank PoC.

Assumptions and constraints, stated honestly. (i) We assume the ambiguous band is narrow and error-dense; if band characterisation in Week 2 refutes this, the architecture's economic premise fails and we will say so rather than proceed. (ii) We assume $q \leq 16$ features retain sufficient band-local signal; measured against a classical model on the same q features, so the feature bottleneck is never silently charged to the quantum arm. (iii) Nyström with $m \ll n$ introduces approximation error; we bound it by reporting spectral reconstruction error alongside downstream metrics. (iv) Hardware noise perturbs the Gram matrix; mitigated below, and quantified rather than assumed away.

Risk	Likelihood	Mitigation / fallback
Quantum arm shows no lift over classical Nyström control	Medium–high	Pre-registered as a publishable result; the band architecture, controls and governance template remain independently valuable deliverables
Hardware noise degrades Gram matrix beyond usability	Medium	Measurement-error mitigation; PSD projection by eigenvalue clipping; ZNE within budget; DM1 noise-model prototyping before any QPU submission
Distillation destroys the lift	Medium	Fall back to batch/offline deployment for flagged-case re-ranking, where exact $z(x)$ is affordable and latency is not binding
Band too wide to be economic	Low–medium	Report band-width/lift curve; select operating point on cost, not on maximum lift
Braket queue latency or device retirement	Medium	Device-agnostic circuits; SV1 results are the primary record, hardware is validation; two candidate devices held

4. Expected Impact

What a successful PoC demonstrates. Not that quantum computers are ready to run fraud detection — they are not, and claiming otherwise would fail the statement's own feasibility bar. It demonstrates something narrower and more useful: *a reproducible, governance-compatible pattern for extracting value from a near-term QPU inside a regulated, latency-bound decisioning stack, with the quantum contribution isolated and measured rather than asserted.* That pattern — expensive expressive model confined to the residual, embedded into explicit features, distilled for serving — transfers directly to AML transaction monitoring, credit decisioning and sanctions screening, all of which share the imbalance, latency and explainability profile of card fraud.

Pre-registered quantitative targets. Stated now so they can be held against us:

Target	Threshold for success	Measured on
Primary — Δ AUPRC vs tuned champion	$\geq +0.010$ absolute, bootstrap 95% CI excluding zero	Full IEEE-CIS temporal holdout
Attribution — Δ AUPRC vs best classical control	$\geq +0.005$ absolute, CI excluding zero	Same holdout, matched protocol
Operational — false positives at fixed recall	$\geq 10\%$ reduction, recall held at champion operating point	Same holdout
Latency — added p99 in distilled mode	< 5 ms	Benchmark harness, band traffic only
Explainability — attribution faithfulness	Deletion/insertion AUC within 10% of champion SHAP	Band sample, $n = 1,000$

Economic translation. We deliberately do not invent a headline savings figure, because it depends on portfolio parameters we do not hold. Instead the PoC delivers a parameterised model, instantiated with the statement's own published constants: for a portfolio of N monthly CNP transactions at base fraud rate r and average ticket A , a reduction Δ in false positives at fixed recall yields recovered revenue $N \cdot \Delta \cdot A \cdot 0.40$ from avoided abandonment (Radial, 2023), plus avoided handling cost, against fraud losses held constant by the fixed-recall constraint and multiplied by 4.41 where they occur (LexisNexis, 2024). HSBC can populate N , r , A and obtain a defensible figure; we supply the machinery and the measured Δ , not a marketing number.

Value even under a negative result. If the quantum arm fails its thresholds, the PoC still delivers: a characterisation of the ambiguous band on the field's reference dataset; a matched-protocol benchmark harness that repairs a real methodological gap in the published quantum fraud literature; the four classical controls as a reusable yardstick; and a documented, evidence-backed answer to "should we put a QPU in our fraud stack in 2026?" A credible negative is worth more to a bank's quantum strategy than an unreproducible positive.

5. Validation Plan

The methodological problem we are solving first. The challenge statement juxtaposes classical baselines measured on full held-out test sets (IEEE-CIS 1st place, AUC 0.9459 on 506,691 transactions) with quantum results such as QSVC $F1 = 0.98$ and hybrid QLSTM accuracy 98.38% obtained on small, frequently rebalanced subsets — in one cited case 35,000 balanced samples. These numbers are not comparable, and the apparent quantum superiority is substantially an artefact of evaluation protocol. The statement asks teams to "clearly report comparison methodology"; we make fixing this the first deliverable, because no claim we make is worth anything until the yardstick is sound.

5.1 Protocol

- **Temporal split, never shuffled.** IEEE-CIS is partitioned on TransactionDT into train / validation / holdout, mirroring production, where the model always predicts forward in time. Random splits leak and inflate; we do not use them.
- **Matched-protocol rule (hard constraint).** Every arm — quantum and classical — sees identical training rows, an identical feature budget q , identical imbalance handling and a comparable hyperparameter-tuning compute budget. Any deviation is reported in the results table, not buried in an appendix.
- **Full-population reporting.** Headline metrics are computed on the *entire* holdout at the natural base rate, never on a rebalanced subset. Band-restricted metrics are reported additionally and labelled as such.
- **One-touch holdout.** The holdout is evaluated once, after all design decisions are frozen. Feature maps, landmark counts and thresholds are selected on validation folds only.

5.2 The ablation ladder — isolating what quantum actually contributes

A quantum model beating a plain GBM proves nothing; it may simply be that *any* band re-ranker with extra dimensions helps. We therefore run five arms, identical but for one component:

# Arm	What it controls for
1 Tuned LightGBM champion, no band stage	The production incumbent
2 Band re-ranking, classical RBF–Nyström, same m , same q	The band architecture itself
3 Band re-ranking, random Fourier features, matched dimension	Simply having more features
4 Band re-ranking, classical deep kernel / learned embedding	A strong learned-representation control
5 QBR — LQKE (treatment)	—

Only the gap between arm 5 and the best of arms 2–4 is attributable to quantum. That single number, with its confidence interval, is the honest headline of the project, and it is the number we will lead with.

5.3 Metrics and statistics

AUPRC is primary, per the dataset authors' own recommendation (Dal Pozzolo et al., 2015); AUC-ROC, F1, precision, recall and confusion matrices at fixed-recall operating points are reported alongside, plus **false positives at fixed recall** as the business-facing metric. Every figure carries a bootstrap 95% CI (2,000 resamples); AUC differences use DeLong; arms are run across ≥ 5 seeds with paired tests and reported variance. An effect whose CI overlaps a control's is reported as null, regardless of point estimate.

5.4 Robustness, hardware and explainability validation

- **Distribution shift:** performance by time bucket across the holdout; by ProductCD and device class; degradation curve as the train/test temporal gap widens. Cross-dataset transfer IEEE-CIS $\rightarrow$ ULB tests whether band structure generalises across portfolios.
- **Hardware:** simulator-vs-QPU comparison of the same landmark Gram matrix, reported as Frobenius distance, eigenspectrum drift, and — the metric that matters — downstream Δ AUPRC. Mitigation stack: DM1 noise-model prototyping before submission, measurement-error mitigation, PSD projection by eigenvalue clipping of the noisy Gram, IonQ debiasing/sharpening where applicable, ZNE within budget. Qubit count, circuit depth, transpiled two-qubit gate count and wall-clock are reported per the statement's good-to-have metrics.
- **Explainability:** SHAP over the augmented vector [band features $\oplus$ $z(x)$]; landmark attribution, which yields the natively interpretable statement *"this transaction's score is driven by similarity to fraud prototypes ℓ_{17} and ℓ_{42} "* — a form of evidence that maps onto how case investigators actually work; faithfulness verified by deletion/insertion curves rather than asserted. Deliverable: a one-page per-decision governance record suitable for model-risk review.
- **Reproducibility:** public repository, pinned seeds, versioned split indices, Braket task IDs recorded so hardware runs are auditable.

6. Hybrid / Cross-Domain Integration

Hybrid by design, not by compromise. Each component is placed where its cost curve is favourable. Classical gradient boosting absorbs volume, latency and calibration — the things it does at microsecond cost. The quantum kernel supplies representation in the small- n , high-ambiguity region where its $O(n^2)$ cost is affordable and its inductive bias is complementary. Classical distillation restores serving economics. Nothing in the design asks a QPU to do work a CPU does better, which is the failure mode of most quantum fraud proposals.

The classical/quantum boundary is drawn at the latency line. Figure 1's dashed line is the architecture's central commitment: every quantum invocation is offline and amortised; the online path is classical arithmetic. Against a ~ 130 ms Mastercard network response inside a 100–300 ms envelope (Mastercard, 2012), no QPU round trip — measured in seconds to minutes on Braket, plus queue — can sit in the authorisation flow. We consider any proposal that places one there to be infeasible, and we have designed to make the question moot rather than to argue it.

HPC and AI integration. Braket SV1/TN1 for landmark Gram computation at scale, DM1 for noise-aware prototyping, local cuQuantum statevector simulation for iteration, GPU-accelerated LightGBM and SHAP for the classical arms, PennyLane for feature-map development with Braket as execution backend. The quantum step is a well-defined, cacheable batch job in an otherwise conventional MLOps pipeline — it produces a matrix, and the rest of the stack does not need to know where the matrix came from.

Enterprise governance fit. QBR deploys as a **challenger** against the incumbent champion — the standard pattern in bank model-risk management. It can be shadow-scored on live traffic with zero production risk, promoted only on evidence, and rolled back by a threshold change. Its explicit feature vector, calibrated probabilities and prototype-level attribution are exactly the artefacts a model-risk function requires. This is what makes the proposal a deployment path rather than a demonstration.

7. Team Capability

Member	Role on this project	Relevant expertise
Kyoung-Tae (Elijah) Kim MSc Quantum Computing, UPM Madrid	Quantum Project Manager — classical arms, ablation ladder, sprint delivery	Project lead, voice phishing (vishing) scam detection, Microsoft; AI & machine learning; RAG and LangChain pipelines; optimisation of agentic AI orchestration systems
Naveed Qureshi MSc Quantum Computing, UPM Madrid	Quantum Technical Lead — feature maps, kernel and circuit design, Braket execution	Quantum optimisation; satellite mission planning formulated and solved as a QUBO
Alejandro Valencia Blancas MSc Quantum Computing, UPM Madrid	Quantum Solutions Architect — pipeline, evaluation protocol, explainability and governance	Software engineering and database systems; mobile communications; artificial intelligence, and the ethics and epistemology of AI

Why this team, for this proposal. **Entangled Minds** formed during the MSc in Quantum Computing at Universidad Politécnica de Madrid, where the three of us worked as a standing project team across the programme's algorithms, optimisation and quantum machine-learning work. That shared grounding matters more than it may sound: we already hold a common vocabulary for encoding choices, circuit budgets and the gap between simulator and hardware behaviour, so a 3–4 month sprint does not open with a team learning how to talk to one another.

The three capabilities this proposal demands map cleanly onto three people. **Naveed Qureshi** has taken a genuinely constrained operational problem — satellite mission planning — and formulated it as a QUBO for quantum optimisation. That is the directly transferable skill here: mapping a messy enterprise problem onto an encoding that fits a finite qubit budget, and staying clear-eyed about when the *encoding* rather than the hardware is the binding constraint. He leads feature-map and kernel design. **Elijah Kim** was project lead for voice-phishing (vishing) scam detection at Microsoft — the closest thing on this team to the problem in front of us: adversarial fraud, severe class imbalance, an attacker population that adapts, and a false-positive cost borne directly by legitimate users. Alongside that he works in applied AI and machine learning — RAG systems, LangChain pipelines, and the optimisation of agentic orchestration. He owns the classical arms and the ablation ladder, the half of this architecture that decides the result: QBR is only as honest as its champion baseline, and an under-tuned LightGBM would manufacture a quantum lift we would not believe ourselves. **Alejandro Valencia Blancas** pairs software and database engineering with formal study of the ethics and epistemology of AI. The engineering covers the pipeline, the reproducibility harness and the serving-path benchmark; the epistemology is less decorative than it first appears, because this proposal's central commitment is knowing precisely what a measured difference does and does not entitle us to claim. He owns the evaluation protocol and the explainability and governance deliverables.

What we do not claim. We bring no prior demonstration of quantum advantage in fraud detection, and we do not believe the published claims on these datasets survive protocol scrutiny — which is exactly why §5 is built the way it is. What this team offers is the ability to construct the honest comparison the field is missing, and to report whatever it returns.

Execution readiness. The team’s AWS account is active with Amazon Braket enabled, so SV1, DM1 and QPU access require no procurement at kickoff. Both datasets — IEEE-CIS and the ULB European Cardholder set — are already downloaded, so Week 1 begins on modelling rather than data access. Classical workloads — the LightGBM champion, SHAP and the four ablation controls — run on free Kaggle and Colab GPU notebooks, which is ample at this data scale and means any reviewer can rerun the classical arms without special hardware. All three members commit full-time (35+ hours per week) to the 3–4 month Phase 2 sprint. The plan is deliberately staged so that Weeks 1–4 produce the band characterisation and the matched-protocol harness — self-contained, independently valuable deliverables — before any quantum result is required. This de-risks the sprint against the single largest failure mode in quantum PoCs: discovering in Week 10 that the baseline was never sound.

References. Deotte (2019), IEEE-CIS 1st place, AUC 0.9459 · Dal Pozzolo et al. (2015), ULB dataset and AUPRC recommendation · Karimi et al. (2024), arXiv:2412.19441, VQC feature maps · Innan et al. (2023), IJQI 21(05), QSVC · Grossi et al. (2022), arXiv:2208.07963, QSVM on IBM hardware · Sakhnenko et al. (2021), arXiv:2112.08869, hybrid QAE · Deloitte + AWS (2024), hybrid QNN on Braket · Nilson Report (2025) · LexisNexis Risk Solutions (2024) · Aite-Novarica (2019) · Radial (2023) · Flagright/PwC (2023) · Mastercard (2012) · AWS Braket device and simulator documentation (2025). Full citations as numbered in the HSBC challenge statement, §7.

Supplementary Material — QBR: Quantum Band Re-ranking

Appendix to the Phase 1 concept proposal. Three items: the Phase 2 delivery plan, circuit-level resource and platform analysis, and full references.

Team: Entangled Minds | **Challenge:** HSBC — Quantum-Enhanced Credit Card Fraud Detection | **Contains no confidential or proprietary information**

A. Phase 2 Delivery Plan (14 weeks)

The sprint is staged so that every gate yields a self-contained, independently valuable deliverable, and so that the project can terminate honestly at any gate rather than drifting toward a result. The holdout set is touched once, in Week 13. Owners: NQ Naveed Qureshi, EK Elijah Kim, AVB Alejandro Valencia Blancas.

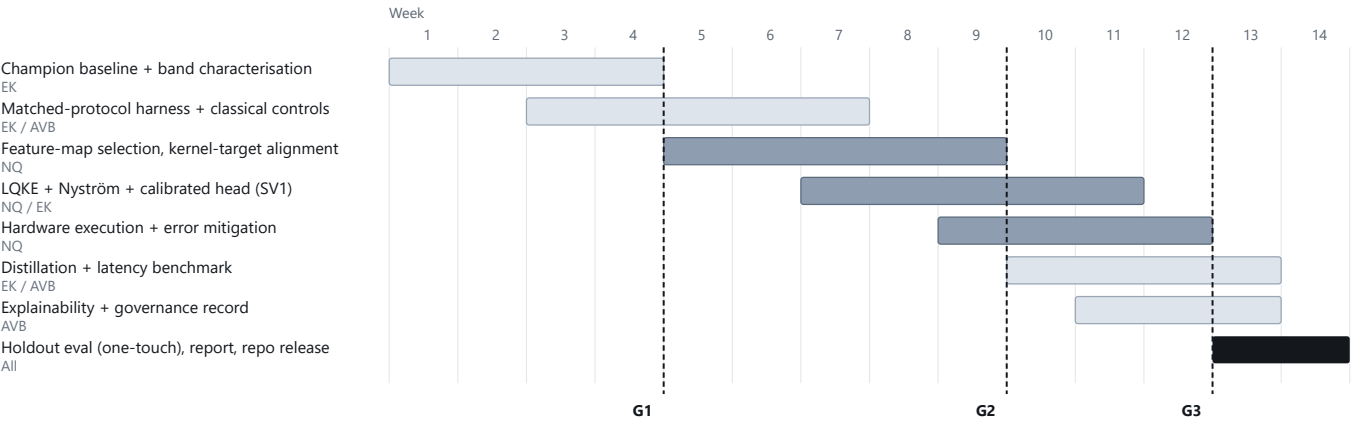


Figure A1. Phase 2 schedule. Darker bars involve quantum computation. Dashed lines are decision gates; each has a defined fail action (below), so the project degrades into a publishable result rather than an overrun.

Gate	End of	Question it answers	Pass condition	Action if it fails
G1	Week 4	Is the ambiguous band narrow and error-dense — i.e. is the architecture's economic premise true?	A band covering $\leq 8\%$ of volume contains $\geq 70\%$ of the champion's residual errors	Stop quantum development. Publish the band characterisation and matched-protocol harness as the deliverable, with the premise reported as refuted.
G2	Week 9	Does LQKE beat the best classical control on the simulator?	$\Delta\text{AUPRC} \geq +0.005$ vs. arms 2–4 on validation folds, bootstrap CI excluding zero	Freeze the design and redirect remaining weeks to characterising <i>why</i> — kernel concentration, encoding choice, or band structure — and publish the negative result with the diagnosis.
G3	Week 12	Does the hardware Gram matrix retain the simulator's downstream signal?	Downstream ΔAUPRC within 30% of the simulator result after error mitigation	Report the simulator result as primary and the hardware run as a noise study, with the fidelity gap quantified.
Final	Week 14	—	One-touch holdout evaluation, written report, public repository with pinned seeds and Braket task IDs	—

B. Circuit Design, Resource Budget and Platform Fit

Kernel entries are estimated by the compute–uncompute (fidelity) method, $\kappa(x,y) = |\langle 0|U^\dagger(y)U(x)|0\rangle|^2$, so the executed circuit is *twice* the depth of a single feature map. All gate counts below are for the full kernel circuit, at $q = 8$ qubits, and assume the standard ZZ-feature-map construction (Hadamard layer, first-order phase layer, then two CX per entangling pair per repetition).

Variant (q = 8)	Entangling pairs	CX in kernel circuit	Survival @ 99.5%	@ 98%	Role in the plan
ZZ, full entanglement, reps = 28 2		224	0.33	0.01	Rejected — depth is not survivable
ZZ, full entanglement, reps = 28 1		112	0.57	0.10	Simulator only
ZZ, circular, reps = 2	8	64	0.73	0.27	Simulator; hardware stretch goal
ZZ, linear, reps = 2	7	56	0.76	0.32	Secondary hardware candidate
ZZ, linear, reps = 1	7	28	0.87	0.57	Primary hardware circuit
Dense-angle (product encoding)	0	0	1.00	1.00	Control only — see note below

“Survival” is the product of two-qubit gate fidelities across the circuit — a first-order planning bound, not a prediction. It ignores single-qubit error, readout error, crosstalk and idle decoherence, all of which make the true figure worse. Measured fidelity against DM1 noise models and against device is itself a Week 9–12 deliverable. Note also that the dense-angle product encoding entangles nothing and is therefore classically simulable; we carry it as an honest control, not as a quantum-advantage candidate, and will label it as such in any results table.

The arithmetic drives a design decision rather than decorating one: **full entanglement is not viable on current superconducting hardware at this width**, and linear entanglement at one repetition is the only variant with comfortable headroom. We would rather discover that here than in Week 11.

Platform fit

Device (Amazon Braket)	Qubits	Connectivity	Published 2-qubit fidelity	Fit for this circuit
IQM Garnet	20	Square lattice	99.51% median	Primary. Linear entanglement maps natively with no SWAP overhead; width comfortable at q = 8–16
IonQ Aria	25	All-to-all	—	Cross-check. No routing overhead even for full entanglement; built-in debiasing and sharpening; higher per-shot cost constrains landmark count
IonQ Forte Enterprise	36 (35 AQ)	All-to-all	—	As Aria, with width headroom for a q = 16 extension
IQM Emerald	54	Lattice	99.5%	Width headroom for q = 16; fallback if Garnet queues are long
Rigetti Ankaa-2	84	Lattice	98%	Width ample but fidelity marginal at our depth (0.57 survival at 28 CX); not planned

Circuit topology is chosen per device rather than transpiled blindly: on a lattice we use the native linear chain, and on a trapped-ion device the all-to-all connectivity removes routing overhead at the cost of slower gates and higher per-shot pricing. Transpiled two-qubit gate counts on the actual device are reported alongside the logical counts above.

Known theoretical risk — exponential concentration. Quantum kernel values are known to concentrate exponentially with qubit count and encoding expressivity (Thanasilp et al., 2024). Where this bites, the Gram matrix collapses toward the identity, the shot count needed to resolve $K(x, y)$ explodes, and any apparent lift is noise. This is the strongest published argument against the method we are proposing, and we would rather name it than be told it.

Three design choices respond to it directly. We keep q small (8–16, not 30+), which is where concentration is mildest. We treat the encoding *bandwidth* — a scaling factor applied to features before encoding — as a tuned hyperparameter, since bandwidth is known to control this effect (Shaydulin & Wild, 2022). And we inspect the Gram matrix eigenspectrum before using it, treating a near-identity kernel as a failure to be reported rather than a result to be tuned around. If concentration dominates at q = 16, that is itself a finding worth publishing, and G2 is the gate designed to surface it.

C. References

C.1 Benchmarks, datasets and industry figures cited in the proposal

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C.2 Quantum machine learning

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C.3 Classical methods relied on by the architecture

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C.4 Platform documentation

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